

ActInf GuestStream 047.1 ~ "Predicting & Reflecting Framework for Dual P

GuestStream | Bellini-Leite | 1:44:44

Hello and welcome. This is ACT-INF Guest Stream 47.1 on July 10th, 2023. We're here with Nick Bird and Samuel Bellini-Lechi, and we're going to have a presentation followed by a discussion. So thank you both for joining. And Sam, thanks again for this presentation. Off to you.

Okay. Was Nick going to say something before I started? Oh, I can go ahead. Okay. Okay. So some introduction first. I'm a professor at Minas Gerais State University. I'm a psychologist, philosopher, mainly working in cognitive science.

I'm going to talk about what I did in my PhD thesis and also some renewed interest I had on this based on some developments in large language models,

which happened this year, most notably Chain of Thoughts and Tree of Thoughts. Those additions to large language model made me want to come back to this again and talk about this.

So how do you know about human reasoning, which is the subject, the main subject underlying this topic?

Well, we had empirical research from the 60s, starting with Peter Watson, who was challenging the notion that we have a logic module for reasoning or some general understanding of logic when we're adults.

And he crafted tasks, which were hard to solve, but they had actually simple fundamentals to it.

Which shows, which were taken to show people's irrationality, something like that.

And also a huge tradition that's called heuristics and biases that also started in the 60s and 70s, mainly by Kahneman and Amos Horský,

which had similar tasks and went beyond and they even won the Nobel Prize for this work.

I had here on the slides evolutionary psychology and language and pragmatics, which were also people in the 90s working with this sort of research.

But I'm going to talk about dual process theory in the way formulated by Evans and Stanovich, which mainly summarizes the results of Watson and Kahneman and Versky,

the Ristic Bias' literature.

So I'm going to talk about how dual process theory came to be and then some problems that it has that I've tried to address.

So one of the first reasoning tasks in this trans literature was the Watson selection task, which goes like this.

Please indicate which cards can determine if the following rule is false.

If a card has a vowel on one side, then has an even number on the other side.

This sounds like a simple exercise.

So the card A clearly is relevant because it has a vowel on one side.

If it doesn't have an even number on the other, then the statement is false and everyone can find this neatly, or most people.

But the problem comes with the card 7, which is not so intuitive.

People actually choose the second one, the card 2 there, which has an even number on one side.

But having that even number doesn't help at all to determine if the statement is false.

What does help is the third, which is an odd number.

And if it has a vowel on the other side, then the statement could be false.

So this was one of the tasks that Watson was famous for.

And Kahneman and Vrowski showed them.

This one, their famous task as well.

It's called the consumption fallacy, the issue that people arrive when solving this task.

So it goes like this.

It is a 31 years old, single, outspoken, and very bright.

She majored in philosophy.

As a student, she was deeply concerned with issues of discrimination and social justice.

And also participated in anti-nuclear demonstrations.

Please make the following statements by their probability, losing one for the most probable and eight for the least probable.

What's interesting here is the statement, the last statement, Linda is a bank teller and is active in the feminist movement.

Linda is a bank teller.

Because there's also these statements without one another, without the conjunction, which is Linda is active in the feminist movement.

Linda is a bank teller.

Because of the description, people think that Linda is a feminist is very likely, given the description.

But they think being a bank teller is not likely.

The issue is that people usually mark Linda is a bank teller and is a feminist as more likely than being a bank teller.

The problem here is that the conjunction of the two necessarily is less likely.

So being a feminist plus being something else or being a bank teller plus being something else is automatically less likely.

So those are the sort of tasks that the reasoning literature used to show some sort of human irrationality.

One issue with this is that it's not very interesting to show that humans are irrational.

It's more interesting to show how humans reason.

And also, the study, these sort of studies didn't show how people actually solved the problems.

They were actually focused on how people made mistakes.

One thing that's interesting about dual process theory is that explains both the mistakes and the solutions to the answers.

So in 2002, Conrad and Frederick formulated this cognitive reflection test, which is another one of these tests.

But it's a lot more simpler and elegant and it really shows, indicates at least how dual process theory is a good solution to a good explanation of the evidence in general.

So, yeah, the Watson selection test used logic and the Linda task used probability.

And here we just have basic arithmetic.

So basic math, we don't have to worry about which answer is truly correct or which rational norms we need to follow.

It's just basic math.

So, there's nothing to discuss here, right?

So the task goes like this.

A bat and ball cost one in 10 in total.

You might know this test.

It's a famous test, even out of the reasoning literature.

The bat costs one more than ball.

How much does the ball cost?

So people, when people read this, they usually say 10 cents intuitively.

But if you take a little time to write it down, you can see that it's actually 5 cents.

So the bat costs one in 5 cents and the ball costs 5 cents together.

It costs one in 10.

So the other two have a similar structure.

The point here is that sometimes we look at these tasks and we have an intuitive answer, which is wrong.

And the way we get out of these issues is that we stop and think a little bit and review our answers with some sort of step-by-step reasoning.

So writing it down, something like that.

What's interesting here is like if you look at 2 plus 2, it's almost impossible not to think 4.

It's almost like you're perceiving the answer is 4 just by looking at it.

But if you have some sort of more complex calculation to do, obviously just by looking at it, you won't have a solution.

So this basic intuition is clearly shown in the cognitive reflection test.

And what's more interesting about it is that the cognitive reflection test actually predicts the performance of various heuristics and biases tests.

So if someone does well on cognitive reflection tests, they're more likely to go well on any other reasoning task.

And if they go bad on the cognitive reflection test, they likely will go bad on other reasoning tests as well.

This is also to me, this is also an indication that this dual process structure of intuitive reasoning versus slow reasoning,

slow reasoning, reflective reasoning, likely explains the rest of the tests as well.

Not only these two, but most of the others.

So it's a good explanation of the literature.

Okay, so what does the theory say besides explaining the reasoning literature?

So in 2008, Jonathan Evans tried to make a formulation of it that would include and unify views from different domains,

such as social cognition, neurosciences, other people in psychologists, which had similar theories. theories.

And it seems like there's something in, in, in, in common here to unify, but Evans wasn't able to.

And so there's a formulation specific for reasoning, and we don't know if it applies so much well to, to other

areas of cognition.

So, one common way to describe dual process theory is by making these tables of features.

So we say, oh, type one processes have these sorts of features on the left, and type two processes have this other type of features on the right.

So this table here was one I made by joining some of the features of Evans and Stanovich, and also of Kahneman.

And then with some questions I, I raised in the thesis, I elaborated it this way, but it's very similar to the Evans and Stanovich formulation.

So, one problem we have with this table's formulation of the dual process theory is that usually people agree that in one thinking instance, when you're solving a problem, you're not going to have all of those features together.

So we don't know for sure when some feature will be present or not, for instance, so clear that type one processes necessarily have to be unconscious in type two processes necessarily have to be conscious.

So, I think that comes a lot because of some conceptual issues in theory.

But one solution that you have found was to have at least some defining features, but different authors have different defining features.

So here I have some of the defining features of the main authors, which is type one processes use less working memory.

But one type two processes most strongly on working memory.

So, type one processes are autonomous in the sense that they don't need to wait for higher processing to respond.

If they have an answer, they go for it.

And using the couples representations mean that is that type two systems can reason beyond what's what's in the present environment.

So, type one processes are not there.

So, type one processes are not there.

Type one processes are faster in comparison, while two is slower in comparison.

Type one processes are less effortful, while two is more effortful in the sense of psychological effort, like people report having a hard time doing things.

So, for instance, in the cognitive reflection test, people that solve the test correctly usually rate it as harder because they had to use type two processing to solve it.

Okay, but Sam knows, which is the philosopher, the dual process theory and other subjects.

He says, Okay, even if this features proposal is good, you see, we still need a mechanism to explain why these features are linked together.

So, what is the mechanism that explains why type two processing goes together?

So, we need some beyond saying that they're different, we need to explain what are these mechanisms that make them different.

And I actually had Samuels on my PhD committee.

And I'm not sure if he was convinced with my solution to my PhD solution to this problem.

I'm not sure how he was convinced of it.

But he approved the PhD, so that's good enough, I guess.

Yeah, the way I...

So, the goal was to solve the unity problem, was to explain the mechanisms that explain type one processing, and explain the mechanisms that explain type two processing.

And I started to see, to search in the literature first, how people were conceiving of the general frameworks that were behind these two types of reasoning.

And one that was not actually related to this literature of dual process theory, but was actually famous in cognitive science was the modularity of mind in the 80s, proposed by philosopher Jared Folder.

And he said that we had two types of processes, a modular process mostly related to input and perception.

And he had central processes that was mostly related to higher reasoning.

So, you can trace some sort of resemblance there to dual process theory.

And the way he went about this is that he said, these perceptual models, modular processes, are information encapsulated.

So, vision only works with vision.

And language only works with language.

And space only works with space.

And so, these were domain specific knowledge systems that did not communicate with each other.

And when they needed context sensitivity, they needed central processes.

So, these central processes, he would argue, were isotropic and quinine, which means that any type of information could be used in a solution.

So, we can relate, I don't know, physics with biology and form a new solution.

So, there's knowledge from multiple domains working here.

And that would also need some sort of context sensitivity, which was a problem for Folder.

He didn't think we would solve this context sensitivity problem at all.

And he had good reasons for that.

In classic AI, we had a lot of issues with that.

I mean, we still do have issues with the context sensitivity problem, but not the sort of issues we had in the 80s when they were debating this same problem of classical AI.

So, this problem was a problem of how to represent a changing environment in symbolic representation.

So, yeah, it was a problem because classical AI attempted to give knowledge to robots or artificial intelligence in statements about the world.

And they were intractable, hard to compute.

There were so many knowledge banks to search and so many knowledge to try to use that the robots couldn't do anything relevant.

They also couldn't represent a change.

So, the changing environment was terrible for them.

Didn't work with this symbolic representation.

It didn't work with serial reasoning or anything like of the sorts.

So, this was a huge problem for classic cognitive scientists.

And since Fodor was a believer in cognitive science in the sense of classical computational theory of mind or language of thought theory, he said that he had found Fodor's first law of the non-existence of cognitive science.

This law states that the more processes are global, the less cognitive science will be able to understand them.

So, he thought only these encapsulated processes could be studied.

And if we needed this sort of global and context sensitive type of processing, then cognitive science just couldn't do it.

Because we had in mind cognitive science in the sense of classical AI.

But it also seems not only Fodor, it seems that the literature abandoned classical cognitive science in various ways, which were strange.

Because if you take the work of Alan Newell and Herbert Simon, it's strange to think that, oh, they were completely wrong.

There's nothing useful that they say that could be applied today.

All the evidence, all the theories they made are just completely wrong.

That's a weird way of thinking.

And, well, if you take, for instance, our theories today of the human Bayesian brain, they don't consider this at all.

So, it's just like they don't get a different strand of theories.

But it always seemed to me that there must be at least something wrong about, something right about this.

It shouldn't be totally abandoned.

There must be a place for these results in the explanation of the mind.

One thing that was in odds with the way Fodor describes his dual process theory, and also sometimes Stanovich, when he talks about dual process theory,

it seems like he is forgetting this fact here, is the fact that we have a limited capacity when we use conscious processing or working memory.

And this is one of the best evidence in psychology we have so far.

We have behaviorist evidence for reinforcement, reinforcement learning and that stuff.

And then we had cognitive psychology in the 60s.

And this is the best evidence that cognitive psychology has had for a long time.

So, it's also something we should not simply ignore.

So, we have in short term memory, the knowledge that when we store some information in short term memory,

we usually can only keep track of about seven items, sometimes even less.

So, in selective attention, there was this task where there was a video of people playing basketball and the subjects had to count how many passes they were making.

And suddenly, a gorilla passed by on the scene and people didn't notice the gorilla because they were so concentrated on the task.

Which is some evidence of selective attention.

That is, when you're concentrated on the task, you actually don't even see other stimuli.

The best overall fit between driving signal and expectations will often be found by, in fact, inferring noise in the driving signal and thus recognizing the stimulus as, for example, the letter M, say in the context of the word mother, even though the same bear stimulus presented out of context or in most other contexts would have been a better fit with the letter N. A unit normally responsive to the letter M might, under such circumstances, be successfully driven by an N-like stimulus. So what this, what Clark is trying to say here that since we have, we represent using probability, then if the context says we should go left, we go left, if the context says we should go right, we go right because the representation

is fluid and it accepts changes really quickly. So that's something that stands from the predictive processing architecture.

But one issue I think that we might see in the predictive processing architecture is the lack of compositionality. So symbolic representations exhibit compositionality, meaning that complex representations are built by combining simpler elements according to rules or syntax. The meaning of a complex representation is derived from the meanings of its constituent parts, which is the way they are combined. This property allows for the generation of new and meaningful expressions by manipulating symbolic structures. So Fodor in 88 with Fulcrum argued that even if connectionless models succeed, they would likely have some sort of simulation of a language of thoughts that exhibits compositionality in order to work like we do. And we see that it seems like it's the case with the results we have this year.

So considering those constraints and the initial view of Fodor and the problem I was trying to solve, which is the unity problem for dual process theory. A major shift I had on the PhD thesis and then I wrote a chapter on this book, the Handbook of Bounded Rationality with Keith. So we explain here in this book the chapter in this shift, the power structure, which is, which is, I mean, it's not clear that, you know, a dual process theories, theorists would adopt Fodor, but it's still very similar.

And then the way the way the way the received view of dual process theory works, which is, oh, type one is a number of dumb heuristics, blind modules and animal intelligence.

And then type two is the real contextual and complex reasoning, the real human intelligence. So I tried to infer that power structure by shape by saying that the contextual predictive processing is, is, is a type one reasoning, which, which is giving us the basics to reason on. So our reasoning already comes with context, we don't need to reason the context with type two reasoning.

We only use type two reasoning to fix some minor issues in predictive processing. And we use a basic search and because it's limited.

We only use it when there's a lot of prediction error and predictions aren't working that well.

And then we call for this realistic search to handle whatever is missing to see if we can find a new solution that wasn't captured.

So in this case, the powerful system is the predictive processing and not the type two research search.

Right.

So that was the general idea for this shift.

Now I'm going to say some hypothesis that stem from this shift.

So T1 processes deal with contents encoded in the form of probability density functions, which is how predictive processing uses to represent the world, which means there is no symbol and no definite content, but values means standard deviation influenced by previous movements in previous world contingencies.

So, manipulating prior information biases the distribution to one or another direction closer to or further from a certain value.

So people here in the active inference institute, we know this by heart.

I'm not changing anything here.

I'm just using what they say basically.

These functions are not stored in the memory bank, but distributed from the responsible brain regions over to the external organs and body parts.

Through neural connections.

The values in the distribution do not represent objects directly indiscreetly.

They refer to distinct aspects of the input when perceptual systems are dealing with such objects.

This is aligned with the T1 processes being easily biased when working with references to similar properties.

Like similar numbers, objects, rhymes, or pet names.

Very often an incorrect value is picked from a distribution.

An example that I have from this is when usually I have seen people confuse their youngest child with their dog.

Dog name with the youngest child.

And when someone has another new child, then the youngest, the new youngest child often confuse the dog.

Because they are, I think, somewhat represented in similar distribution, similar points of the distribution.

This is also in line with claimants of embodied proposals that the world is not represented in symbols.

T1 processes are sub-personal and their predictions are made by the same systems which process perception.

So, a clear example is that a judgment about a facial expression is related to the FFA, which is a region of the brain that processes faces.

And this is also related to that judgment that Kahneman made, the argument that when looking at someone, we make judgments about their face.

The idea is that perception is not passive, but already comes with predictions.

And when in problem solving, such prediction is precisely the type one answer.

So if we look back at the reasoning tasks, all those type one tasks are likely stemming from predictions.

I don't know what to claim that C1 processes are purely perceptual if in context to cognitive, only that such predictions stem from perceptual processes.

So point in fact one reasoning is that there's not a clear line between what is perception and cognitive.

So I think the word you just, using different words for this is the problem.

So just not getting into that.

Kahneman's example of judgment of negophageous expression shows how this is expected of dual process theory.

Like it's expected of dual process theory that type one reasoning works under predictive processing.

Also in line with the claims of embodied cognition that there is no sharp link between perception and reasoning.

On the other hand, T2 processing works like a classical machine for reasoning, such as general problem solver of Newton Simon.

Part of this classical machine only makes sense in the brain if it exists in a wider set up of predictive processing network generating type one responses.

Like in Newell's physical symbol system, when facing reasoning problem, T2 processing opens a problem

space containing an expression that designates the initial problem.

And an expression designates the solution, which was produced by probabilistic prediction.

Having the initial expression in the predictive expression in the problem space, T2 processing then uses the smooth generators to attempt to reduce differences between them and sometimes find different solutions in such path.

So let me just explain this a little bit better.

When Newell was working in the 60s there, they knew they couldn't search all possible space of a broader space because it wouldn't work.

So they had this, had this, a holistic search, which worked like a detective.

Let's ignore the problems which are likely wrong and just look at the spaces that might be right.

Let's ignore the problems.

What this predicting and reflecting framework does for a holistic search is that the risk search only works on, it works already with the prediction that would stem from type one processing.

So it's like we only start reasoning and making searches based on the prior predictions we already had.

Right. And we all, we only go further with the search and the predictions aren't working.

And the main point here is that the risk of search isn't searching a random space or even a pre-programmed space.

It's searching a space that was left out from the predictions that started in type one processing.

So that's how it doesn't go to the frame problem anymore because

a large part of the contextual issue was already solved by predictive processing.

Okay.

So

that was the general hypothesis I presented to attempt to solve the unity problem, which is the problem of how type one features go together and how type two features go together.

And now I'm going to try to argue that this is likely the case.

So I'm going to try to argue that this is a good reason for this is a good reason for us to think that type one processes

use predictive processing and that type two processes do not use predictive processing.

Rather, they need some kind of symbolic, a realistic search.

So I'm going to try to argue that this is a good explanation for the difference between implicit representation and explicit representation.

If you look at the literature in psychology,

you will see that it's very ambiguous in these definitions.

Like people use implicit and explicit representations as the same thing as unconscious, conscious or fleeting, graspable or the same thing as automatic in control.

And so when it goes to dual process theory,

it's not very clear what they're saying in each case, but they use this expression implicit and explicit a lot.

How I think this model solves this is by saying that implicit representations are probability density functions, while explicit representations are symbolic representations, classical symbolic representations.

Clark, in his book, sometimes mentions that we can have single peak distribution distributions, but the problem with having single peak distributions is that, well, if you only have one peak on the distribution function, then it's not. It doesn't have all these features of a realistic representation. It's more likely similar to a symbolic representation. In fact, that's having a single peak distribution might be the way that a probability representation is turned to a symbolic representation. These have different and important features. So the probability representation is continuous. It's uncertain and ambiguous. And that's why it's able to be sensitive to context. In the example, the mother example of Clark. It can vary on retrieval. We can remember something and be different the second time. It's tied to priors. It works with statistical relations. Yeah, it's called in and depending on which value is most probable at the time, it has a different outcome. So I argue that explicit representations are likely discrete and they're likely symbolic in nature. They're unambiguous. This is what allows us to disapplicate in the first place. They're stored reliably. If you have a reliable definition of it, you will likely remember it the same way the next time. It's arbitrary in the sense that it's not really coupled to the stimuli in the stimuli in the sense that the perception, our perception usually is. The statistical relations are related to aspects of the world. It's using compositionality and it's immutable in the sense that you can't change it. You can't change it. It's fixed. This value doesn't change. So that's the implicit versus explicit feature. You can have also explanations for automaticity in contrast to working memory. So automaticity concerns over-learned skills and over-learned skills are understood as skills that have become predictable in this framework here. So we usually use this to explain like driving, riding bicycle, riding. So when you're driving, if you don't know how to drive, you need certain statements like how you should steer, right? You need to have those in mind.

But then once you learn how to drive, it's like your body already knows what it's doing and you don't even need to think about this.

So this is a classical distinction in psychology.

But I argue that the predictive processing theory makes a new enlightening to this.

So it's like our body is predicting what we're doing.

So it's automatic.

So it's an explanation for automaticity.

But say that a dog certainly does suddenly runs from the car.

You obviously don't have a model for that.

So you're going to need to call in working memory to solve some of those issues.

So you don't hit the dog.

So everything works under prediction unless the model is very unreliable.

And then working memory is called in to solve some further issues.

An interesting hypothesis I had on the PhD thesis, which I never explored.

And I actually don't even think anyone read this part.

But it seems to me to be a very interesting hypothesis in relation to free energy and active inference.

So that's why I'm here in the Active Inference Institute.

That as we know, the brain attempts to minimize free energy by getting predictions right.

So the next slide does, higher need of type two processes are related to higher free energy and to create information in the sense that when the predictions are working clearly, then you don't need to have effort.

But when there are a lot of prediction error, then you need effort because you call in the working memory to do a research search.

So to minimize a great amount, better amount of free energy will take more time in work.

This is more effortful than having predictions ready that minimize free energy as quickly as possible.

Before when probabilities fail, the system needs to start investigating possibilities.

It searches for other possible solutions by means of the risk search.

The risk search will be related to more information and time because it does not have probable solutions ready. Instead, it needs to investigate its state space almost from scratch.

We say almost because it is the risk enhanced. It also will have tricks to get the correct solution faster. Unlike group force search, which would investigate the state space from beginning to end.

We believe reports of effort would be related to executing more risk searches.

This reports of effort by subjects would seem to be based on cognitive, informational and physical constraints of reality. So that's why I think this is an interesting part of the hypothesis.

It's because we have this psychological measure of effort, but if this is true, then we actually have a

physical measure of effort. Like if people are working hard on the problem, that means there's a lot of free energy going on and they are having to use this research search to fix some issues that the model isn't able to fix by itself.

Yeah, so that's the effort part. So now going for the working memory part, what's interesting here is that in working memory,

working memory, it's like it's expected of it that it works like a classical computer.

working memory. So working memory is a widely researched topic in psychology, but in predictive processing,

it's rarely, rarely mentioned. And I actually searched the predictive processing books with control F for working memory. And people rarely use this topic because it's not done by predictive processing. It has nothing to do with predictive processing. It's essentially an issue for predictive processing.

Because, well, if you're going to eliminate classical symbolic processing altogether, then we need a predictive processing explanation of working memory. And so far, I haven't found one. And I don't think there will be one. But this also could be just ignorance of mine. Maybe there's some paper there, which I haven't found.

Might prove me wrong. But in any way,

working memory is very aligned with what Turing was thinking when he was making the connection between machines and minds. So if this is a sentence of Alan Turing explaining the his computer, right?

The Turing machine. And if we change the word computer to working memory, it clearly works. It's almost like it's

talking about the same thing. When, of course, if we change the word here, computer to predictive processing, it's clearly not talking about the same thing.

So the behavior of the computer or working memory at any moment is determined by the symbols, which it's observing in his state of mind at that moment. We may suppose that there is a bound beat to the number of symbols or squares, which the computer can observe at one moment.

If he wishes to observe more, he must use successive observations. We will also suppose that the number of states of mind, which can be taken into account is finite.

The reason for this are the same characters as those which restrict the numbers of symbols.

If we admitted an infinity of states of mind, some of them will be arbitrarily close and will be confused.

Again, the restriction is not one which is seriously effective computation since the use of more complicated states of mind can be avoided by writing more symbols on the tape.

So this is very similar to what we think about when we're talking about working memory.

And not only because we were influenced by Turing, but because of the evidence that comes from working memory is very similar to what's expected of this sort of machine.

So this is very similar to what we think about.

Finally, we have speed.

So predictive processing has various strategies to make the processing be as fast as possible.

While that's not true for symbolic classical AI.

So yeah, Clark says cheap, fast world exploding action rather than the pursuit of truth, optimality, deductive

inference is now the key organizing principle when he's talking about predictive processing.

The predictive processor is always taking certain bets about what the current state of the world implies, losing accuracy and compensation for speed.

We also have predictive coding in the more strict sense.

So by predictive coding, we mean specifically the property of the system to consider from the world only stimuli which result in greater prediction error.

So there's also a filter there which allows for speed and perception.

And focusing on predictive prediction relevant stimuli only permits the agent to quickly decide forces of action and select amongst the possible forces.

So predictive processing is tailored to be fast necessarily.

Well, that's not the case for symbolic AI.

And so that's the explanation for the difference in speed.

So beyond that, the tip two processes need to figure out the solutions online.

So it's different if you have a prior prediction that will say to you what the answer is from having to search a space from scratch.

Right.

There's also another issue that the biological brains certainly were not built for a serial heuristic search.

And Fodor and Pilschens say that the moral is that the absolute speed of a process is a property for excellence of its implementation.

And since the brain does not have a symbolic processor implemented, if we do use some sort of heuristic search, it's like a different adaptation.

It's not it's not what the brain is used to.

And searching problem spaces is slower than having a problem outcome ready in the system.

Okay.

So that was the work I did in my PhD thesis.

And those were one of the main reasons why I think this framework is good for explaining the differences in features of dual process theory.

And I actually wrote a prediction that I just I recently learned that I wrote this because I forgot I wrote this.

And then I went back to the thesis and I said, well, I actually said this would happen.

And it kind of did.

So that was one of the reasons why I renewed my interest in my PhD thesis.

So what I said was predictive processing system would be subject to bias from lack of compositionality, such as mistakes in transitivity, failures in noticing necessary character form rules and so on.

Precisely the type of mistake type one processing in person.

So at that point, we didn't have large language models.

In fact, the attention is all you need paper came out in the same year.

And then I noticed that this could be seen as a prediction of the type of issues that large language models are facing.

That's reliability, hard time keeping the order of units or steps in complex reasoning or math straight, hard

time letting go of priors.

And to my.

What really motivated me to start working on this again was that they solve this precisely by using dual process theory in T2 agents that are going.

So what they're doing now to solve these reliability issues is adding a risk search to the genetic models and it's it's having awesome results.

This is just this year.

It's just in 2023.

So the more they develop these type two agents to answer these.

Reliability problems of.

Large language models, the more the large language models are getting good at stuff.

So this is this is a paper I got.

This is the images from the paper.

Tree of thoughts, right?

This one, which is one of the last ones.

Probably came out about two months ago.

Something like that.

The way they're doing this implementation is they're taking the.

The results outputs of the large language model and they're doing further reasoning on that.

And then feeding the large language model with the result of the further reasoning, the further reasoning they're doing is very similar to the classic symbolic AI.

So here they explain there's a simple input output prompting here, which is simply inputting a prompt and the large language model will give you an output.

And as we know, these these outputs sometimes are very biased.

There's hallucination on it.

They're not reliable.

No, all that stuff.

And last year they figured out that through chain of prompt chain of thought prompting, we could also already increase.

The reliability of this, these models.

So you give an input and just as we do the better prompting, you know, when you go to chat to PT and say, I'll do this better fix this issue.

Solve this step by step.

And some of the commands that are good for chapter chat to PT and chain of thought prompting automated some of that.

So those reasoning steps and then large language models were working much better after this.

So this was last year.

And then last year this started a new trend of research to to make these agents coupled to large language

model output.

So this is a new trend of research to put linguistic prompts back into the large language models to see how they can get better.

The self consistency one, you give the input and then there are various solutions.

The self consistency forces the large language models to try different solutions and then you get a majority vote.

So if you have like five similar answers in some different paths, then this one's picked over the one which is less popular.

And the tree of thoughts, which I thought was most similar to the proposal I had in psychology and the process theory.

Is the one which opens up the problem space.

It's actually pretty much what I just explained in the case of psychology.

They open up the problem space and they start to search possible solutions which are better than the original.

And they feed that back into the predictive processing.

Oh, sorry.

The generative model, the large language model.

And then this one is currently one of the best ones they have.

So here I made a general diagram of how this would work.

It's not psychology nor AI.

It's like what's similar in the two.

So we have a generative model here using probability density functions.

This looks more like a traditional cognitive connection is modeled in a predictive processing.

It's just it's just the diagram is not that good.

I'll talk a little about that.

And then.

So, yeah, we generate an answer here, which is to the problem here.

And we get a linguistic output.

And this linguistic output then goes for an heuristic search using symbolic representations.

So if you get a better answer here, it feeds it back to the generative model and so on.

Basically, the knowledge bar is the generative model.

And the heuristic search is simply using more steps to see, to prompt it back.

It's more like a prompting scheme than a new knowledge scheme, which is both true for the dual process theory model I created and the heuristic search that's happening here in the T2 agents for LLMs.

A better way to do this, and I think Tree of Thoughts does implement this, is that the input layers is trying to predict the thoughts, right?

It's a bit more like predictive process than the other diagram.

And so they're trying to predict the thoughts and by open up this, searching this problem space during these steps, we already have new prompts here for the generative model.

So if you take predictive processing seriously, and this does happen in the brain, it's more likely like this. I do think that the Tree of Thoughts paper does say that the intermediary processes interfere in the searching on the elements.

Finally, I thought maybe we could suggest something from psychology to this model to keep keep growing, working differently based on what we know from psychology. So some things that are not implemented yet, that executive functioning have these updating abilities in intelligence forgetting, which is related to insight problem solving is when you you forget something, but that it means like you forget prior, right? You're using a different prior, so you you're able to let go of biases, right?

There's also work on thinking dispositions, which could be relevant.

Thinking dispositions are sentences like this.

Police should always be revised in response to new information of prior evidence.

So this would be like imperative linguistic knowledge, which would have to be added on.

I don't know where, but likely would help work.

There's a lot of thinking dispositions, thinking dispositions that are relevant to our successful reasoning. And this is just one example.

Yeah, when the literature knows a lot more examples that could be relevant.

We have in creativity research, we have this generative phase, which is similar to what the generative models are doing, but we do not have the exploratory processes going on, which is common in the creativity research literature. But it's what we do with, for instance, mid-journey when we're doing better prompts.

So maybe using these exploratory processes from the creativity research, we can also automate the exploratory part of the creativity processes that these image generators are getting to not just the generative part.

And finally, I think the embodiment, we don't have robots or anything like that that do this sort of reasoning.

So AI on the computer and also stuff from active inference, which was not considered slightly related to embodiment in the sense of navigating the world and predicting the world. So that stuff is far from happening in the traditional LLMs we have.

So yeah, that's a lot of stuff.

Maybe you guys have some comments.

As I said, I haven't been able to talk about this to anyone.

I do think I have some, at least some, maybe some have relevant stuff that I've published, but since I'm not famous and I don't have people to talk to, it's mostly gone unnoticed.

So thanks for the space here and time.

Thank you.

Thank you.

Awesome presentation.

All right, Nick, it'd be awesome to hear your introduction and then take it wherever you'd like to go.

Sure.

So first, thanks, Samuel, for interesting talk.

And I've been really pleased to find your research.

I guess

maybe I'm supposed to say something about who I am where I am.

So I'm Nick Bird.

I'm at the Stevens Institute of Technology in the New York City metropolitan area in a department that's kind of interdisciplinary.

So we have like philosophy and quantitative social science, but also like people doing like music and visual arts and all sorts of other things.

I tend to do more of the philosophy and quantitative social science stuff.

So I'm kind of also interested in human reasoning.

And so much, much of the people I'm reading and citing and drawing on in my research are a lot of the people that you saw in the opening slides that we saw, you know, people like Wason and Kahneman, Tversky and Mercier and Sperber.

And a lot of the other people that were cited, I think Samuel does a really good job to incorporate the kind of like shoulders of giants that we're standing on in terms of the philosophy of mind, right?

The voters and the politicians and these people.

And one thing that I think is, you know, on everybody's mind these days is these large language models.

So I'm really glad to see how this this dual process theory that was that emerged from research like Kahneman and Tversky's on these these problems like, you know, the conjunction fallacy problems or the Wason selection card task or the conjunction fallacy tasks, how that has been applied to models like large language models and what we can learn from that.

And then I think the coolest thing about the preprint print is how we could kind of port back and forth the learnings from both the computer science and the cognitive science to improve one another.

Right. So I think that one of the neat things about this preprint that Samuel was giving us at the end is that, you know, we're kind of taking what we think is a model of reflective reasoning that we like, you know, we represent certain parts of a problem and maybe reason more carefully and effortfully about them in certain situations in ways that might help us.

And how can we help these large language models do the same?

Because it does seem like large language models function.

At least the ones that we interact with online mostly, they seem to function mostly as like a system one or type one process, they're just kind of like quickly generating lots of text or imagery or something like that, but they're not necessarily like reflecting on it and the ways that we would think that a, you know, a human is capable of.

And they might not even be capable of like representing things in the same way that we do.

Yeah, so I thought that was a really interesting idea, this using chain of thought and tree of thought models to

kind of create a reflective level or reflective system within the models.

It seems just like a really valuable idea and just a really great synthesis of research in multiple disciplines.

Something that I think few people are actually very good at is like incorporating some of the best insights from multiple fields were often pretty siloed in academia.

So I just think this is great work and I hope more people will appreciate and pay attention to it and build on it.

So one of the things that stood out to me in this presentation more so than when I was like reading this 2023 preprint analytic reasoning for large language models by Dr. Bellini-Leiche.

The thing that stood out to me is this idea that like Bellini-Leiche and Frankish sort of switched which of these two types of reasoning processes are supposed to be context dependent or context sensitive or however you would want to word that.

And I started thinking, well, I wonder if there's a sense in which both types of responses or both types of reasoning are somewhat context dependent.

And I'll just say what I mean and then maybe Samuel you can kind of say what you think I should think or clarify the view of the predictive and reflective framework or something.

So the thought I was having was, well, in a familiar context, these intuitive predictions, these type one processes, our gut response, so to speak, those are going to be pretty useful because they're well trained.

We have a lot of experience that we're drawing on.

So like our gut response, our first response is often quite good.

It's in these like less familiar contexts or maybe similarly familiar, but like way higher stakes or something.

But it's in these other contexts where we might think, oh, maybe I should slow down and make sure I've double checked whatever my initial impulse is before I just accept it.

Because there's a lot riding on this or like I'm just not used to this type of problem.

So I need to, you know, I need to slow down.

And so there's a sense in which what context is doing is not just showing up in one or the other type of reasoning, but it's sort of like determining which type of reasoning might be best at the moment.

But I'm wondering, like, is that just totally compatible with what you're imagining?

Or is that like somehow a deviation from the framework, the predictive and reflective framework?

Yeah.

Thanks for the thoughts there and compliments.

And thanks for the question as well.

So the reason why I think context needs always some sort of type one help is because of the constraints we have learned from symbolic AI.

So if I'm saying that type two reasoning have the same constraints, then it cannot be contextual because it doesn't work like that.

We know that if it fails in context, it fails bad.

But of course, we do need to like when we have a novel situation, we often solve this by context.

Right.

So how could this be?

And I think it's by the interconnection of the power of the two systems.

So we always start with the prediction, which in this case would be mistaken.

And then we start, we have to search for novel instruments.

And in this search, we make the contextual comparisons.

So it can't be that the system two is making the comparison on its own.

I think it's more likely that when it's a case of context that stems from a novel situation, and then the two systems have to figure out it together.

But likely, in the sense that I was pointing out here in the end, these predictions that go over the thoughts, they most likely are the ones that will solve these context issues in novel problems.

Yeah, I'd argue something like that.

Okay, I think that's somewhat helpful.

So then maybe what I'm thinking is something along what the lines you were saying, at the end, you were saying how there's still kind of an opportunity to understand things like executive function in this framework.

And I think that's maybe part of what I'm wondering about.

So I guess I'm wondering if, yeah, if there's more to be said within this predictive and reflecting framework about how each of these two types of processes get selected or like, what might help the system?

Can I hold on to that specific point you just remembered?

I just remembered something that's related.

So yeah, clearly, our working memory or type two reasoning likely does more than that.

It does this, but likely was more than that.

So there's there's likely also a belief bank, something like the thinking dispositions, some sort of knowledge bank, which is related to type two processing.

So Keith Frankish does this distinction between flat out beliefs and.

I'm not remembering the terms, but he has a type one belief in type two belief.

So type one belief is that belief you have, but you're not fully confident of it.

And the type two belief is that belief you have when you you have a, say, a very firm political position and you state it with obvious words that you are sure you believe that.

So there's likely to be some sort of different belief structure as well, which I don't talk about at all in the thesis of the work.

So I'm not saying obviously that this is a complete model of reasoning.

And obviously we need more to figure out how reasoning works.

So, yeah.

Yeah.

Okay.

That's that's that's also helpful.

I'm wondering if Daniel, did you want to weigh in?

Well, I think it's a very interesting question about which of those modes are engaged or how they're balanced through time.

And how does this connect to what's known as the context window in today's transformer type?

Large language models.

so how do you map the computational attributes of large language models today like architecturally or their ram or cpu usage how do you map that onto for example human cognitive processes do you think that's useful or that there are any insights there well i'm mostly making a relation between predictive processing and generative models but i know that i mean large language models are not entirely the same as what active inference is saying i'm aware of that but there is some similarity specifically in regards to the generative model parts so i would i'm not sure i'm not a

a.i specialist to to be sure which details of a generative model could be true of the human brain and i'm most likely betting that the the people who study generative models in the brain like the the fristons and all have figured out something like what our generative models do one other thought or nick do you have a you want to add there um just thought of some different ways whether people have connected it explicitly to the literature on working memory in predictive processing i i think as you pointed out it's definitely a link that is not highlighted it's one one that we recently heard um from uh professors walker and uh manriquez and friston in the recent live stream 53 that was on like cognitive paleoanthropology looking at human working memory but some ways that

people have incorporated working memory is like a nested model that carries context at a deeper time but that is not necessarily like an actual mechanism that gives rise to why type one type two are the way they are um so i thought that was a very provocative direction to go to move past the descriptive and then to look at the underlying generative model of why these outcomes are the way they are even though these cognitive phenomena are causes of other things happening they are also caused by some influenceable um or contextual or variable aspects and when we lose that context dependence of the cognitive systems then they're totally lifted and disembodied don't really help

so by putting the the primacy on that predictive processing element i think it opens the door to connecting those areas beyond just mentioning the terms in proximity but to really support different epistemologies from predictive processing as a model or approach

yeah can you go can you go on that idea again i'm not sure what you want me to comment can you summarize that again and make it a question sure what do you think the epistemological consequences are of taking predictive processing the way that you have approached it here versus alternative

or or prior approaches to cognitive sciences

okay but what do you mean by epistemological

uh

uh

formative how how we know advanced with this theory

um how it influences our our understanding of how we know or seek or practice or do or decide uh

perfect okay so let me let me think about that for instance um yeah i think one of the best uh uh what this helps with most is explaining dual process theory because dual process theory is in that shape in terms of of concepts uh in terms of theory and formulation uh

people have no idea what these these systems refer to in the mind brain uh they often have you know different intuitions into this and basically basically what i most think this work is relevant for is for saving dual process theory from criticism uh which has been happening in psychology so people are like oh it seems like you're talking about the left and right brain something like that something that we we don't know for sure what these two systems are right um

so this is a good way forward i think uh for for dual process theory and as uh said the process theory does have some value in itself uh so yeah saving the process theory i think is a good uh direction for for this this model but also if we if we think about the advantages of it would be something like for instance it could have helped us had the idea of using chain of thoughts or or tree of thoughts had anyone read this before so it can help us make these these these sorts of predictions that we shouldn't have some sort of zero reasoning coupled to the generative models so yeah i think it could help us think on further on on ai if we uh yeah like like like like nicholas saying to like incorporating more stuff in psychology that we know that's certainly missing here but here we have a direction of what we should incorporate and and so on yeah but yeah i'm also aware of the working memory research from predictive processing you mentioned and if you want to go back to that and explain me some more of that i'd be happy to hear yeah sure so not that this is the most effective way to implement a memory system but at least that this provides uh analytical method for measuring or describing them if one we're going to have the five digit number you can imagine a a nested model where the lowest level is the ones place and it's nested within a decision tree of tens places and hundreds and so on and then there's some seek or access policy or strategy that helps speak the number in reverse and some cognitive or computational limitation is just how well that cognitive agent can perform on that test again that doesn't mean that's the mechanism that's being used but that would be a way to use a nested generative model to encode like multiple levels of spatial or temporal variability but context becomes an issue because you're basically just expanding the the possibility space looking for sparser and sparser associations so if you don't have a good compositionality or really well definable well articulated causal process then you're just building these all-by-all models that you're going to be searching through in the dark yeah one point i want to comment on that is that well it's not clear that the best way to reason will be the way humans reason right so we might eventually have a different way of implementing a working memory on an llm which would be more effective than ours there's also to the other side there's also the point that it seems that the limited capacity we have on working memory maybe it's not just a lack of power maybe it's necessary so we kind of like don't lose our minds in the sense that we have a limited space to search and then we're able to search the this limited space and if it was too uh if you have if we had too much space to search because we had a super super working memory then we wouldn't be able to finish the tasks anytime so maybe this is like a physical good uh barrier that's something that helps and it's interesting that what i find interesting is that okay the the ai may find a lot a lot of different better ways to implement your work memory but it's interesting that they implemented a working memory or a search agent which was very very like the way i described in the thesis that that was the what i found amazing as you're saying there's there's there's very ways to implement they work in memory but they did it in a very similar similar way in the sense of searching the space after the generative model uh has offered a solution right so i don't know maybe that's a a good a good uh indication that this is on the right track but it also may just simply be a simple way they

found to to to solve the problem and it was just by coincidence similar to this so yeah maybe the correct way or the way the brain does may also be different we don't know right all we have was the the limited evidence i i offered and a bit more obviously and uh some constraints you know on ai and information processing and you have to figure it out and i think that this implementation stuff we're you're asking is mostly mostly due to the ai people to find out or the the math people in this instance to figure out

nick do you want to comment or a question or i'll ask one from the chat

yeah so i just have a kind of a bigger picture question that kind of gets us into the realm of these chat bots um right so you in this preprint and um in this talk have like given us a variety of different tasks in the heuristics and biases literature the the conjunction fallacy to uh ask the the linda problem some people call it um or these cognitive reflection test questions and um you know there's a there's like a paper in the fall that showed that a preprint uh in the fall that last fall that showed 2022 that showed gpt3 and 3.5 and even the earlier versions like showed a lot of the human-like faulty intuitions like it basically fell for the lure pretty often um sometimes like most of the time it was falling for the lure but then as soon as chat or gpt4 uh was was available to be studied and used on these models it was like performing basically near perfect and when it did get the incorrect response it wasn't necessarily the lure it was just like some other general type of error it was making like misinterpreting the question altogether or something and it seems like the predict the predicting predictive and reflecting model framework could have at least two different ways of explaining this like one is like uh what happened is they changed the ai's like type one thinking or whatever so that like it could uh respond intuitively to all uh to all these but intuitively and correctly so it didn't need to reflect or it could be that like they somehow created this like reflective type system with either like you know a chain of thought or a tree of thought or some other type of system that helps it actually engage in reflection on these these tasks or whatever the analog of reflection would be for a chatbot um and so i'm wondering like if you have thoughts on this like as you've been thinking about this or talking to anyone about this like which how do you think they did this obviously we have to speculate because a lot of the data is proprietary but i'm just curious to get your thoughts yeah that's an awesome question i actually forgot to mention this on the presentation that the uh the the biases they make on the reasoning test last year were exactly the ones the the humans made so if we three uh fed it some dual process real it would give exactly our mistakes that's that's so amazing i i'm not certain why that happened but it did and um how they fixed this so yeah i do think that they trained the model on these tasks because people were able to like uh get similar similar issues by changing the questions a little bit so i think they like cheated on that in the sense that they trained on unknown problems but that's not true of new problems if you should give it some different formulation and feed it to i don't know 3.5 new version it's still it's still failing but uh the people on open ai they they are aware of this chain of thought research they are aware of thought research but i don't think they want to depend or rely on these external agents so they recently uh uh released a report where they are trying to reinforce intermediate

steps of reasoning on jpc4 so they're trying to simulate internally what the the the chain of thoughts reasoning is doing externally by reinforcing the intermediate intermediate steps of reasoning so that it stops uh failing composition helping so on okay i'll add a thought on that i was reminded by daniel dennett's framework or model in darwin's dangerous idea of cranes and skyhooks and so some building is built and then the question is like was it built with cranes or with top-down hung skyhooks that just sort of descend from nowhere hold everything up while it's all being built and it's like well no it's with cranes and then if you need a big crane you can use a smaller crane to assemble a crane so that's kind of a bottom-up constructive metaphor whereas the skyhook is like the top-down compositionality and that's when you get to a certain level of sophistication for a cognitive system it can make the blueprint or the plan like maybe what we would associate with the expected free energy calculation not just the variational free energy calculation so at some point when a plan can be made it requires a strategy between type one and type two switching and probably other switches too um and so that crane approach you need a lot of compute to do a bottom-up so even though it's kind of weird to think about it's almost like the current large models are very bottom-up because they're very bottom-up from syntax and they don't have um what you refer to as uh imperative linguistic knowledge which can be normative as well as um a heuristic for knowledge and that's potentially a small model but it contains wisdom that's explicit and practices and um um stances or or dispositions uh as you uh as you described and so it's kind of like it's not a denial of embodiment to also clarify what this skyhook ability is and it's type 2 like so i think um i guess what do you think about that or how do you connect this to anything in that area yeah i'm gonna make i'm gonna make you do the summary and the question again just although i didn't track some of what you mean i'm not sure which ones mean to comment on that do you think that the uh landscape of large and small models would be different if people took on board some of the the features you described like imperative linguistic knowledge seriously into construction of models so let me just see if this if this is something you're asking so they're they're doing small models now for phones and open source models for phones and they have the issue of not being able to scale scale up right because if you scale up too much then you can't run the phone uh so you're asking if we can implement something like a tree of thoughts to to help these small smaller models is that is that interesting yes are you sure or just are you just saying okay go for that that's an example of what it would mean to re-understand like where do we need an api call to a cloud center or where could some context be locally computed with much more of a type one what are some type two problems that we might be able to type one our way out of what are some type one areas where things are not working and a type two um description might be advantageous yeah yeah i think that they're they're getting on to that and they're they're being able to make these smaller smaller models precisely because of these sorts of um discoveries they made on chain of thoughts right uh there's also stuff about being able to train on specific data uh i read one i think it's called textbooks are all you need they're the training smaller smaller models on textbooks and they're doing better so yeah there's there are differences in the training training smaller models in different data but there's also this the possibility of

making them more reliable based on what i'm calling t2 agents or train of thoughts stuff like that and again we don't know if that's all we can do externally but i would argue that we would need necessarily some sort of external mechanism coupled to any large language model for the best results because of the differences in representations mainly because of this right unless the large language model somehow uh makes a symbolic representation emerge from its processes it's not clear that it has done so but unless it's it does so

and it uses that reliably because it may even have generated some sort of symbolic process internally but it doesn't always use that or doesn't know when to use that so yeah most likely we're going to need something external to the language model and um yeah this this is some clues to to how to implement to implement that and also i think in our brain it's kind of external not only in the sense of our minds but also in the brain since it's a recent step in evolution uh we don't see i don't know and lizards reasoning in type 2 manner right uh we do see chimpanzees really in type 2 manner in type 2 manner but not as much as we do so it's very human-like reasoning uh it's just that going back to the context problem uh animals do solve contextual problems that that's why they can they can live right so the issue is not actually the context there it's it's more like uh some some boost on reasoning capacity and that boost on reason capacity may be related to the prefront the prefrontal cortex as we know and so of course the prefrontal the prefrontal cortex is also a network it's not a serial tuning machine but if uh it's it's there's somewhere that's implementing a serial machine or a symbolic classic AI machine or symbolic reasoning something like that most likely it would be the prefrontal cortex which is kind of like a distinct addition an addition to the brain it may be something related to language results which is also a distinct a new addition to the brain so yeah and also i'm very convinced by the photo and pollution argument in 88 that okay you're gonna need symbolic processing anyway right either the model is gonna generate that symbolic processing or you're gonna have to feed it for any because or else you're gonna have hallucination or mistakes in reasoning because of the nature of uh because of the nature of um distributed representations right it's it's uncertain and ambiguous by nature that's why i'm not entirely convinced that only probabilistic representations would be enough to solve hallucination issues

yeah

awesome nick do you have any other questions or areas you want to mention or ask

i think um i think those were my main questions um i don't know if there's anything in the chat that you wanted to touch on before we go i know we're maybe over time

i'll go with one from the chat and then we can have any closing thoughts so glia maximalist glia maximalist asks are you familiar with visual pathways in the human brain specifically the dorsal where and ventral what pathways do you think this maps onto the two systems you described for problem solving okay okay yeah so like i said in 2008 evans tried to accomplish that unifying vision of dual process theory but i'm not very uh convinced by that i mean he he also said well we can't do this as of yet but i i do think it's possible as i said on the presentation i just don't know how to do it uh how to have a um convincing theory of everything of dual process theory in the brain uh this it does go on to other

regions like you have automatic motor responses and controlled motor responses uh and and like you said there's a different different pathways in reasoning uh in figuring out stuff in in the uh visual field so yeah maybe the what would be related to something and where to another most likely the what would be would be uh tapping on to type 2 processing i'm not sure how the where would be tapping on that but yeah as as i'm trying to make the point if we extend it to the whole brain it we lose what we we've learned we're not sure how that applies cool well nick any penultimate last words

but you muted

oh wait unmute and then continue sorry about that um thanks uh thanks to both of you

uh dr bellini light and uh daniel for coordinating this and i'm really just glad to have found the research and to discuss it and connect and uh looking forward to further conversations to be honest awesome sam any closing thoughts yeah i'm just sad i couldn't go much uh beyond what i said in the the presentation yeah most of the stuff you you guys uh asked uh i'm not sure i have good answers for that basically what i've done is all i know and i'm not sure how to go uh

on the future with this and maybe someone in the audience audience would do so i hope that that helps that helps great i hope so too all right until next time thank you bye

um